

Myke-Vital Temgoua Sao
Department of Physics, Faculty of Science
University of Yaoundé I
P.O. Box 812, Yaoundé, Cameroon
myke-vital.sao@facsciences-uy1.cm

August 28, 2026

Prof. Kenneth M. Merz, Jr., Editor-in-Chief
Journal of Chemical Information and Modeling
American Chemical Society

Dear Prof. Merz,

We submit our manuscript **“Interpreting Docking-Derived Resistance-Retention Scores in Antimalarial Triage”** for consideration as an Article in *JCIM*.

This work evaluates a docking-based resistance-aware prioritization strategy and uses targeted molecular dynamics as a separate structural follow-up. It does not assign a systems-level mechanism of action from docking alone. Seventeen Set-C candidates were ranked by a docking-derived resistance resilience score (RRS) across six PfDHFR and PfCRT resistance states, with chemical-space (ACSI) and network (PNS) scores kept as separate ranking dimensions. Four parent-study complexes that are not on the Set-C list were simulated for 10 ns each as a short pose-retention check. On the 12 candidates with complete two-target data the classes were one A*, one A, four B, five C, and one D; five PfCRT-only compounds are reported separately. Only PfCRT-214 gave an interpretable MM-GBSA estimate (-18.25 ± 0.40 kcal mol⁻¹). An external docking panel (39 ligands, 312 records) averaged near RRS 100, with little mutant/wild-type contrast under the same protocol. A Tartarus cross-check showed no composite MPO-affinity association, consistent with treating those scores as separate ranking tools.

A 16-system Set-C MD pilot on two candidates tests whether short trajectories and MM-GBSA recover the docking-RRS pattern. It is a single-replicate structural stress test, is not used to assign RRS classes, and is not biological validation. Sensitivity checks explore candidate removal and score perturbation. Companion Paper 1 results provide upstream methods and chemical-space context only, not independent replication.

Code, source data, and analysis scripts are available on GitHub under an open-source licence; a Zenodo snapshot will be deposited at publication. Statistics are reported with effect sizes, permutation *p*-values, bootstrap intervals, and Bonferroni adjustment.

Suggested reviewers.

1. Dr. Fidele Ntie-Kang, University of Buea — African natural-product computational discovery
2. Dr. Kelly Chibale, University of Cape Town — antimalarial drug discovery
3. Dr. John D. Chodera, Memorial Sloan Kettering — free-energy methods and MD best practices

This manuscript is original, not under consideration elsewhere, and all authors have approved the submission.

Sincerely,

Myke-Vital Temgoua Sao
(on behalf of all co-authors)